6.1 Derivative function and standard functions

Key point · Definition and results. A function is differentiable on an interval when it is differentiable at each of its points; its derivative associates with x the number $f'(x)$. The standard derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)' = nx^{n-1}$ on its domain of differentiability; for $n = 0$, the derivative is 0 directly.

Status and justification. Square: the rate is $2a + h$. Reciprocal: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{[\frac{1}{a+h} - \frac{1}{a}]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are accepted here.

Example 1. For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f'(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function exists at 0, its standard derivative only for $x > 0$.

Watch out. The domain of differentiability may be smaller than the domain of definition.

Application. Differentiate x^5, x^{-2} and $\sqrt{x}$, stating the domains.

6.2 Sum, product, reciprocal and quotient

Key point · Definition and results. For differentiable u and v, $(u+v)' = u' + v'$, $(\lambda u)' = \lambda u'$, $(uv)' = u'v + uv'$. Where $v \neq 0$, $(\frac{1}{v})' = -\frac{v'}{v^2}$ and $(\frac{u}{v})' = \frac{u'v - uv'}{v^2}$.

Status and justification. For the product, add and subtract $u(a+h)v(a)$ in the numerator. The rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. Since a differentiable function is continuous (accepted), we obtain $u(a)v'(a) + v(a)u'(a)$. The other rules are accepted.

Example 2. For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f'(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping the brackets avoids changing the sign of a single term instead of the whole numerator.

Watch out. The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

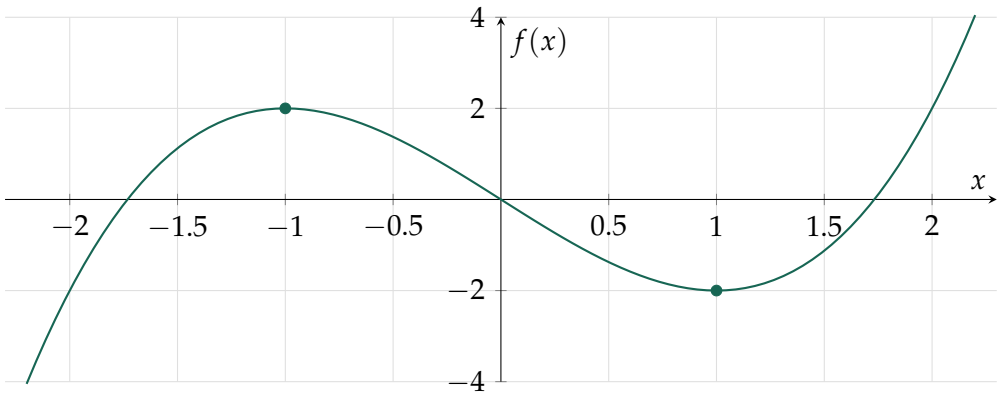
Application. Differentiate $(x^2+1)(3x-2)$, then $\frac{1}{x^2+1}$.

6.3 Sign of the derivative and extrema

Key point · Definition and results. On an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, it is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies strict increase. If f has a local extremum at an interior point where it is differentiable, its derivative is zero there. The converse is false.

Status and justification. Monotonicity theorems are accepted. For an interior minimum at a, the difference quotients on the right are nonnegative and those on the left are nonpositive; if they converge to the same real number, it is 0. The same argument applies to a maximum.

Example 3. $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f increases on $(-\infty, -1]$, decreases on $[-1, 1]$, increases on $[1, +\infty)$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. Neither of these extrema is global on $\mathbb{R}$.



$f(x) = x^3 - 3x$: a local maximum at -1 and a local minimum at 1.

Watch out. The theorem applies on an interval. A function can decrease on two separate intervals without decreasing on their union.

Application. Study $f(x) = x^2 - 4x + 1$ on $[0, 3]$ and give its extrema on this interval.

6.4 Parity, inequalities and comparison

Key point · Definition and results. On a domain symmetric about 0, f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$. The curve of an even function is symmetric about the y-axis; that of an odd function about the origin. To compare f and g, we study the sign of $f - g$, possibly with its derivative.

Status and justification. The symmetries follow from the coordinates of the corresponding points on the graph. To prove an inequality, studying monotonicity may give a minimum of the difference and hence its sign; the derivative does not directly replace this difference.

Example 4. For $x > 0$, $h(x) = x + \dfrac{1}{x} - 2$ has derivative $\dfrac{x^2 - 1}{x^2}$. It decreases up to 1 then increases; $h(1) = 0$, so $x + \dfrac{1}{x} \geq 2$. Equality is obtained only at 1.

Watch out. Check the domain before studying parity. The sign of $f' - g'$ tells us about the monotonicity of $f - g$, not directly about its sign.

Application. Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

6.5 Optimisation

Key point · Definition and results. Optimising means defining the variable and its domain, expressing the quantity to optimise, then comparing all the candidate values, including the endpoints.

Status and justification. For a function continuous on a closed, bounded interval and differentiable in its interior, the minimum and maximum occur at endpoints or at stationary points. Compare the function values at these candidates.

Example 5. A box has a square base of side x and height $6 - x$ (the sum of the side and the height is 6), with $0 \leq x \leq 6$. Its volume $V(x) = x^2(6 - x)$ has derivative $V'(x) = 12x - 3x^2 = 3x(4 - x)$, positive on $[0, 4]$ and negative on $[4, 6]$. The maximum volume is $V(4) = 32$.

Beyond the syllabus · Newton's method. The tangent at x_n meets the x-axis at $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ when $f'(x_n) \neq 0$. For $f(x) = x^2 - 2$ and $x_0 = 1$, we obtain $x_1 = \dfrac{3}{2}$ then $x_2 = \dfrac{17}{12}$, whose squares are 2.25 then about 2.00694: we approach $\sqrt{2}$. Convergence is not guaranteed in general. This method is not part of the première syllabus and is not assessed.

Watch out. A numerical value produced by an algorithm is not a proof of a minimum or of convergence.

Application. A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation.